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From colour target to ICC

InkProf - Quality profiling for RGB printers

Requirements and startup

Source version 1.4.2, released 2026-10-27. Tested with base MATLAB R2025b (no additional MATLAB toolboxes) and Python 3.13.6 on macOS. The macOS program folder includes separately licensed ArgyllCMS 3.5.0 tools for Apple Silicon and Intel. MATLAB is installed separately; setup_python_environment prepares the program's own pinned Python environment and needs a suitable Python 3.13.6 interpreter. Python performs analysis and reporting; MATLAB provides the app and workflow. Windows and Linux are untested and require compatible tools; a POSIX bridge does not establish Linux qualification.

How to use this guide

This guide uses row for a numbered entry in the application's left-hand workflow table. Run selected step is the app's button label. Follow saved status and suggested navigation; a completed calculation is evidence, not automatic approval of print quality.

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One project, one source of settings

Use "New project" once, then "Open project" to resume. "Project details" contains the "Project and materials", "Printing settings", "Profiling" and "Target paper" tabs. Record printer, paper, ink/coating, application and driver settings, drying time, profile name, colour-data choices and paper/sled limits. Unknown values remain unknown. Shared project-controlled recipe fields are read-only in the recipe dialog. The separate "Compensate to D50 (also updates Project details)" FWA checkbox remains editable when spectral data are available; saving the recipe also updates the project’s FWA setting. Relevant changes invalidate dependent results while preserving earlier evidence.

Rename through "Project details" to rename the project folder too. External folder renaming is detected for confirmation. Move the entire project folder between computers; locally discovered Python and Argyll paths belong to the computer, not the portable project. "Verify project" checks recorded file hashes. "Close app" after saving/completing the active operation; reopen the same project after drying.

FWA and measurement conditions

FWA compensation models D50 effects from suitable M0 spectra. It does not convert an M0 instrument measurement into a native M1 measurement. Record the actual instrument conditions and compensation used. You can enable compensation later without changing code: retain native spectra, save a new B2 recipe, rebuild B3 and repeat checks. New physical verification is required for the changed profile.

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Workflow rows 1-10

Row	Action	What it creates or decides
1	Create/import RGB target	RGB definition and generation metadata
2	Save TIFF16	Matching TIFF16, TI2, layout and manifest; print separately
3	Measure/select revision	Instrument readings and saved JSON/TI3 revision
4	Review and accept	Explicit acceptance of the selected revision
5	B1: lock inputs	Immutable, checked profiling input snapshot
6	B2: recipe	Manual build recipe; project settings are reused
7	B3: build ICC / automatic iteration	ICC candidate, build arguments and job log
8	Fit, Grid and C1	Model fit, transformations and numerical checks
9	C2a. Prepare verification print (TIFF16)	Profile-applied verification target
10	C2b. Measure verification print / select saved measurement	Saved verification revision

Build and check labels

B1 freezes validated input; B2 records the build recipe; B3 builds the ICC. C1 checks numerical transformations, inverse behaviour and CMM consistency. C2 produces a profile-applied print target; C3 compares its measured colours with desired colours and profile predictions. The Grid check samples an RGB lattice and checks round trips and ramps. Automatic iteration builds and evaluates candidates under its recorded selection rules; it does not approve a profile.

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Workflow rows 11-19

Row	Action	What it creates or decides
11	C3: compare model and print	Desired vs measured and predicted vs measured errors
12	Review feedback	Recorded feedback/decision, not merely opening a window
13	Approve for intended use	Open review and approval opens the dialog; Approve for stated use saves the reviewed decision
14	Save ICC and measurement certificate	Delivery copies plus PDF/HTML measurement certificate
15	Refinement tools	Image, manual gamut/shell, errors or ICC-grid trials; saved refinement target
16	Measure refinement	Reviewable additional measurements
17	Build next iteration	Combined training set and a new ICC candidate
18	Compare previous/current ICC	Both ICCs evaluated at common RGB sample points; not dependent on their measurement grids
19	Save without print verification	Numerically checked later iteration and explicitly limited certificate

Ready: Required inputs are available. Select the step and "Run selected step" to start.
Complete: The step’s result or decision is saved and the app’s requirements for that step are met. This does not itself mean approved print quality.
Locked: The step cannot start yet. Read its description and first complete the prerequisite steps.
Stale: An earlier result needs updating because its inputs or settings have changed. Repeat the affected steps.
Use the dialog’s intended save or acceptance button. Simply opening and closing the dialog does not complete the step.
"Refresh" reloads saved status; it does not save edits still open in a dialog. Finish or cancel the active dialog before starting another operation.

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Targets, paper and measurement

Print the matching target

TIFF16 is saved, not sent to the printer. Use "Copy TIFF16 for printing" and read the accompanying PRINTING.txt. Print at 100% under the recorded conditions and allow drying. Base and refinement targets contain device RGB: disable further application/driver ICC conversion. C2 colours have already been converted once through the selected ICC; do not convert them again. Device-RGB reference patches in C2 retain their specified values. Keep matching TI2 and layout data in the project. Patch-row labels appear on both sides; compact layouts reserve a footer band.

Choose paper dimensions

"Target paper" suggests cut sheets A5 through A3+ or roll/custom sizes according to patch count and equipment limits. The 320 mm scan-width and 370 mm length defaults describe one sled, not universal instrument limits. Edit the limits in "Project details" and review the proposed layout.

Measure and review

Follow calibration and measurement prompts. Instrument identity is recorded when available. A remeasured patch remains a candidate until accepted and saved as a new revision. Select that revision and repeat dependent reviews/checks. A Lab value describes one measured colour; a change in Lab compares two readings of that patch. Optional ColorChecker reference targets can support comparison with familiar reference colours; their presence depends on the selected target and recorded evidence.

Instrument instructions

See docs/usage/instrument-startup.md and docs/usage/chart-measurement.md for calibration, reconnect and retry instructions.

Save and resume completed rows

Choose Save completed rows / pause and wait for confirmation before closing. A save requested during scanning waits for a row prompt and excludes the unfinished row; paired rows require both scans. Later choose Resume saved session and its folder, then Resume measurement. The dialog states page, row and direction. Confirm the same printed sheets; reprinted or replaced pages need a new session and fresh RGB checks. A checkpoint is not a completed workflow measurement. Real-instrument resume validation remains necessary.

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Refinement and shadows

Open row 15, "Refinement tools", to choose the source of additional patches.

Use image colours or C3 feedback to propose more patches. An image's embedded ICC is used; if absent, a warning precedes the sRGB fallback so you can choose another image. Estimated-error filtering defaults to ΔE00 > 5. These are predictions, not measured errors.

Unmeasured C2 colours can be included with the new target. C2 colour/grey/challenge patches train the next iteration; repeated and paper-white patches retain control roles. Once used for fitting they are not independent validation of that next ICC. Challenge patches deliberately probe difficult colours or transitions; they are not automatic proof of print accuracy.

Project details > Project and materials: Paper finish = Matte. Profiling: Matte shadows = Extra shadow detail for Matte; Dark patch emphasis = 2; Shadow grid emphasis = 1.3; Extra iteration patches = 48. New refinement targets reserve about one third of the shadow budget for neutral patches and one third for weakly coloured shadows (predicted C*ab up to 10); remaining slots cover other dark colours. With 48 available patches this normally gives 16 per group. Dense device-grey probes and small channel offsets supplement Argyll sampling. Unused quota slots return to the dark pool. Existing RGB16 colours are excluded. The receipt records policy and source ICC hash. Print device RGB without colour conversion, measure the new TI2 and build the next iteration. The current ICC only predicts selection; it is not applied to this print. Original targets and profiles remain unchanged. Verify the candidate with a fresh print; more patches cannot guarantee a darker physical black.

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Shell refinement and continuation

Use Refinement tools to strengthen coverage near the gamut surface.

Review before printing

Choose the whole inner shell or selected areas, set geometric thickness in ΔE76, and review plotted candidates. Save the TIFF16 package and use its matching target for measurement. Saved packages are registered in row 15 without regenerating patch order.

Continue from measurements

Measure in row 16 and build in row 17. The build runs internal Fit, Grid and C1 diagnostics. Registering the new iteration marks the workflow checks stale and selects row 8; run row 8 to record the checks for the new active iteration before verification or delivery. Row 18 compares profiles. A build stopped by numerical alerts retains measurements and the previous active ICC; row 17 results link its latest C1/Grid reports.

Choose the delivery path

With current numerical checks, choose either rows 9-11 for a new verification print and measurement, row 12 to review feedback, row 13 to approve intended use and row 14 to deliver; or row 18 for comparison and row 19 for explicitly numerical-only delivery. Earlier measurements are historical evidence, not verification of the new ICC.

Iteration: 16 -> 17 (internal build diagnostics) -> 8 (current workflow checks) -> 18 (comparison)
Print-verified delivery: 9 -> 10 -> 11 -> 12 -> 13 -> 14
Numerical-only delivery: 18 -> 19

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Interpret and deliver

Fit error against training data, numerical consistency and physical print verification answer different questions. Large desired-colour errors with small model-vs-print errors may reflect gamut limits; this alone does not prove the cause. Shared-sample profile comparisons show change, not which profile is more accurate. Compare in 2D starting at L*=50, or switch to 3D. View gamut shows a separate ICC-derived surface; HTML can rotate it, PDF shows a still view. BPC (black point compensation) is not applied to the diagnostic absolute-colour comparison.

Stable delivery names and HTML support

The project retains the checked ICC candidate. The proposed external filename combines the profile name, iter-N and the final eight characters of the iteration UUID; the save dialog lets you choose its destination. The same stem identifies the ICC and certificate. The delivery ICC description is updated, so its file hash differs from the checked candidate even though colour-tag payloads are preserved. Keep the complete delivery folder and its supporting evidence folder (underlag/) for portable HTML, linked reports and assets.

Certificate and separate measurement appendix

The main certificate contains scope, project/printing conditions, ICC identity, summary, available ColorChecker reference/measured pairs, decision and signature space. Historical ColorChecker results are labelled as such. Complete accepted measurements and technical details are in the separate measurement-appendix.pdf. Appendix A and Appendix B inside the certificate explain methods/standards and legal terms; they are distinct from that separate PDF. The measurement appendix is linked by certificate ID and the checked candidate’s SHA-256. The certificate separately records the named delivery ICC and its SHA-256.

Interpret standards carefully

Selected ISO 12647-7:2016 comparison limits are referenced through MediaStandard Print 2018, published by Bundesverband Druck und Medien (bvdm). This custom workflow is not ISO certification. Numerical-only delivery explicitly records the absence of separate print verification.

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Code, licensing and runtime checks

InkProf code and authored documentation use GNU GPL v3 or later, without warranty; see LICENSE and THIRD_PARTY_NOTICES.md. The macOS distribution includes separately licensed ArgyllCMS 3.5.0 tools for ARM and Intel, with upstream notices and provenance. MATLAB and a portable Python runtime are not bundled; the isolated Python environment is prepared locally. Dependencies, schemas and fonts retain their own terms. See docs/releases/v1.4.2.md and LICENSE_SCOPE.md for current scope. The review in licenses/review-v1.0.0.md is historical evidence, not a legal review of the complete current release.

Version checks and reporting problems

InkProf checks required tool versions and Python imports during setup. Incompatible versions or unavailable imports can prevent operations. Use the setup diagnostics and saved job logs when reporting a problem through the public repository’s Issues page. This guide does not promise a support service.

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Identify the printed target before measuring

A saved date alone is not enough. Match the target type, page count and original package path to the physical sheet.

Base target or C2 verification?

Row 3. Measure / select measurement revision reads the base target saved in row 2. A C2 print belongs to C2b. Measure verification print / select saved measurement. If its package is already selected, go directly to that measurement step; do not generate the target again. After all rows are read, choose Save and finish, then C3. Compare model and print and Review feedback.

Target locked is intentional

A prepared measurement is bound to its TI2. The disabled Target locked button indicates that target selection is fixed. To change the target, close the measurement window and select the matching target in the workflow. Do not substitute a similar-looking chart while a session is prepared or partially measured.

What the window shows

The summary shows patch count, target type when recorded, page count, saved local time and UTC. Target details shows the full path and offers a page preview. Missing historical metadata is not guessed. Compare these facts with the printed footer. Package creation time does not prove the physical print date or drying time.

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Readable print folders and traceability

Copy TIFF16 for printing creates a separate delivery with the project name instead of a UUID-only folder.

Folder naming

Example: My printer paper_iter-1_c2_20261024-123456. The prefix is the saved project name, followed by the workflow iteration at copy, step and local copy timestamp. A collision creates another numbered folder rather than overwriting a previous delivery. This is the project name, not an independently entered ICC/profile description.

PROJECT_INFO.txt

The UTF-8 text file retains the exact project name and ID, original project path, workflow step, current iteration and iteration ID, target-origin iteration and ID when found in history, UTC and local copy/creation times, target description when present, source TI2 and its SHA-256, project printing settings and copied filenames with SHA-256. Unknown origin metadata stays unknown.

Copies versus originals

PRINTING.txt retains printing instructions. print-delivery.json carries the structured receipt; the workflow also records the copy in project reports/history. TIFF data is copied and checked byte for byte. The physical sheet is not identified merely by a hash: match its footer and documented settings. The TIFF-only delivery does not replace the original TI2/layout for measurement.

A target can outlive an iteration

A base or C2 target may have been created in an earlier iteration. The folder names the workflow iteration at copy; PROJECT_INFO.txt separately records the target-origin iteration recovered from matching history. Package creation date and copy date are different events and neither proves when the sheet was physically printed.

Complete target-package copy

The optional complete-package copy from refinement tools uses the same project-based naming and project identification when its source is inside a recognised project. Without project context it uses a generic readable print-copy name; missing project/iteration facts are not invented.

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Verification and complete delivery

C2a prepares the print and C2b measures it

C2a. Prepare verification print (TIFF16) saves or selects the print package. C2b. Measure verification print / select saved measurement reads that sheet. Choose Measure for a new print, not Resume saved session. Save and finish, then run C3. This verifies the existing ICC; it does not build a new profile.

Read C3 feedback

Run Review feedback to open a readable feedback window. Reopen it with Open selected step results while feedback is selected. C3 files are verification-check.md and verification-check.json under targets/<target>/checks/<check>. Each check has its own folder. Compare reports with the same ICC and target; distinguish error against desired Lab from error against the ICC prediction.

Approval and certificate are separate actions

Selecting a row navigates only. Run Open review and approval, review the results, tick the confirmation and save your assessment. Then select row 14, Save ICC and measurement certificate, and click Save ICC and certificate... Choose Also save copies elsewhere to select the bundle name and location. Save in project only keeps the delivery inside the project. A previous numerical-only export from row 19 does not prevent delivery after print approval.

The exact verified ICC is included

The external delivery folder contains <profile>-verified-original.icc, a byte-identical copy of the checked ICC. The normally named ICC has updated name metadata; an optional -v4.4.icc is a converted variant and is not the original verified file. delivery.json records verifiedOriginalICCFile and verifiedOriginalICCSHA256. Keep the whole bundle, including the supporting evidence folder (underlag/).

Repeat verification after changing printer settings

Print the same target again with the intended settings and document the actual Chroma Optimizer (CO) or other coating choice, media, quality and drying time for each sheet. Start a new C2b measurement; previous measurements remain. Compare matching C3 reports. A saved Automatic setting alone does not establish the physical coating used.

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Current workflow: decisions and navigation

Review and lock once

Row 4 keeps the measurement preview open until Continue to assessment is chosen. B1 validates the selected revision, opens a modal Lock profile input confirmation, then saves a checked snapshot. Activity and elapsed time are shown during validation and saving. Suggested navigation is B1, B2, B3; B2 is shown for recipe review, while Automatic B3 still creates its own recipes.

Patch counts and result windows

Extra refinement patch parameters apply when adding patches or continuing an iteration, rather than during the initial B3 build. C1 saves its numerical reports without opening successive result windows automatically; use Open selected step results. C2a handles verification-target preparation; choose existing files when reusing a printed sheet.

Correct recorded details

When Project details changes, distinguish Correct recorded details from Changed actual setup. A documentation correction requires a reason and unchanged physical conditions. Original file hashes are checked and a new audit record stores before/after values. Numerical processing changes cannot use this exception. Existing stale steps are not automatically restored.

Measurement uncertainty and verification

Repeated measurements show measurement spread. Review numerical alerts and assess results with an independent print. An absence of observed problems does not establish that noise propagation through refitting has been fully studied.

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Current workflow: measurement and delivery

RGB and other spot measurements

RGB controls, single-patch replacements and paper-white spots use a short start sound for screen-triggered and i1Pro 2 instrument-button measurements. Calibration does not use the spot start sound. RGB colour rows show page, colour and progress only. The temporary sound-test button has been removed. Page changes are shown inline without a separate confirmation window.

Cancel and resume safely

Skip / close and the window close button can cancel the RGB control dialog while calibration is waiting. Saved requests, transcripts and readings remain. Partial output lines do not block GUI polling, and Argyll debug text after a prompt is handled. Main workflow activity is cleared on interruption. Restart InkProf after source changes; resume saved chart rows only for the same physical print.

Refinement print copies

After successfully saving and registering gamut patches from Refinement tools, the workflow gamut window closes. The standalone viewer also closes after successful TIFF16 creation. To copy an existing refinement target, select row 15 in the main table and use Copy TIFF16 for printing..., then choose Downloads. The new folder identifies the project and iteration and contains printing files and traceability information. The source project is preserved.

Timed certificate delivery

Row 14 shows activity and elapsed time while collecting evidence, rendering figures and PDFs, preparing ICC variants, and copying and verifying delivery files. The external bundle includes the exact checked ICC as <name>-verified-original.icc as well as named delivery variants. A stored certificate is not regenerated in place; create a new export revision when changes are needed.

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Gamut boundaries and extended targets

Explore the current model boundary

View gamut offers Gamut boundary exploration (both sides), alongside the existing inner-shell and selected-area methods. Set Boundary distance in ΔE76, minimum RGB spacing and maximum new patches. Candidates are printable device RGB, not invented out-of-range colours. A negative signed distance means predicted outside the current mesh; it does not prove physical gamut expansion.

Extend latest target

Tick Extend latest target to combine the latest completed gamut print for the exact current ICC with new points. Unprinted proposals are skipped. Previous fitting/development points and supplementary shadow points are retained; exact duplicates and existing training RGB are excluded. Controls are regenerated. Saved measurement rows or readings block extension of that target. Opening the measurement window alone does not block it. The earlier TIFF, TI2 and readings remain unchanged.

Print and measure the new version

The complete target is saved normally in the project. The optional external-copy dialog defaults to Downloads; Cancel skips only the external copy. Print every page without colour conversion. Row 16 measures the new TI2, and row 17 builds the next profile with the earlier training. Do not measure an earlier layout with the new TI2. View gamut closes after successful creation and the copy dialog; cancellation/error leaves it open.

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Review measured Lab density

Read the measurements

Refinement tools > Review measured Lab density / balanced trial lists close measured Lab groups and their RGB span. Readings is an integer count of repeated RGB16 observations; maximum ΔE00 describes their repeatability. Similar Lab at different RGB may describe saturation. No original measurement is deleted, overwritten or averaged.

Set, recalculate, select

Adjust Lab distance (ΔE76), Target per group and Repeat review (ΔE00), then Recalculate. Defaults 1 / 4 / 1 are provisional. Use balancing applies automatic representatives to checked groups; unchecked groups keep all rows. Select all checks/highlights all groups; Clear all restores every row to the proposed input. Neutrals, lightness/RGB extrema and inconsistent repeats stay protected. The target count may therefore be exceeded.

Build a separate candidate

Close only closes the review. Build separate balanced candidate uses the reviewed subset and is disabled when zero points would be omitted. Try different settings and Recalculate rather than forcing removal. Both profiles are compared to all original fitting observations, including omitted rows, to available reserved development patches and to matching photographic gradients. These are diagnostics, not final independent print verification. No candidate automatically replaces the active profile.

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Activity, provenance and computer changes

Visible work stages

Density review, Recalculate and the balanced trial show elapsed time and current phase: input verification, original snapshots, ICC build, all-patch comparison, original/candidate gradients, available development checks and result saving. Changing settings requires Recalculate before building. Trial parameters, checked groups, source rows and SHA-256 hashes remain in separate review/trial records.

One environment per program

Each program has its own .venv with Python 3.13.6 and the same pinned requirements. From the code folder use setup_python_environment("check"), "sync" or "rebuild". A rebuild preserves the previous environment as a backup and restores it on failure. Do not commit or move a .venv as a portable replacement for rebuilding it. Projects and measured originals are separate from the runtime.

Native Argyll and documentation scope

The program folder includes separately licensed ArgyllCMS 3.5.0 for ARM and Intel macOS, with upstream notices and provenance. Native tools are preferred and checked for actual startup. This document describes release 1.4.2. Restart the app after source updates.